

YieldFlo User Manual

YieldFlo is a yield monitoring application that works alongside AgOpenGPS (AOG). It records yield rate, moisture, and GPS position continuously while harvesting, and produces spatial yield maps and job reports.

YieldFlo receives GPS and section control data from AOG over a local network (UDP). A YieldFlo hardware module connects to the combine's clean-grain elevator and moisture sensor and sends live readings to the PC application.

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1. Getting Started

Requirements

Item	Requirement
Operating system	Windows 10 or Windows 11
.NET Framework	4.8 (pre-installed on Windows 10 May 2019 Update and later)
AgOpenGPS	Running on the same PC or local network

No installer is required. Extract the YieldFlo folder to any location and run `YieldFlo.exe`.

Hardware module

- YieldFlo module (based on ESP32) mounted on the combine
- Connected to the clean-grain elevator optical sensor
- Connected to the capacitance moisture sensor (optional)
- Communicates with the PC via WiFi or CAN bus

Network

- **WiFi mode:** connect the PC to the same WiFi network as the YieldFlo module, or directly to the module's access point
- AOG must be running and broadcasting GPS/section data on the local network

Recommended first-run setup

On first launch YieldFlo creates a default crop, header, and profile. Before starting your first job:

1. Open **Menu** → **Settings** and select your preferred units (Imperial or Metric)
2. Open **Menu** → **Profiles** and enter your combine ID
3. Open **Menu** → **Crops** and add or edit crops for your operation
4. Open **Menu** → **Headers** and enter the correct cutting width
5. Open **Menu** → **Fields** and add field names if desired

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2. Main Screen

The main screen displays live harvest data and job status.

Data panels

Panel	Description
YIELD	Instantaneous yield rate (bu/ac or t/ha), smoothed over recent readings
MOISTURE	Live grain moisture percentage from the capacitance sensor
SPEED	Ground speed received from AOG GPS
Totals area	Total area harvested, total mass, and average yield for the current job

Sensor bars

Two vertical bar gauges on the right side of the screen:

- **Left bar — Elevator flow:** Shows the grain obstruction percentage in the clean-grain elevator. Higher = more grain flowing.
- **Right bar — Moisture:** Shows the relative moisture sensor reading.

The numeric values below each bar show the raw sensor counts from the module.

Status bar

Indicator	Green	Orange	Off / Red
GPS	AOG connected and sending position	—	No AOG data
Module	Module data received within 5 s	—	No module data
Job name	Job recording (active)	Job paused	No active job

Buttons

Button	Action
Menu	Open the main menu
Start	Start or resume the active job
Pause	Manually pause recording
Stop	Stop and close the current job
Exit	Close YieldFlo

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3. Jobs

Menu: Menu → Jobs

A job records yield, moisture, GPS position, and area for a single harvest session. Each job is linked to a crop, header, and profile.

Starting a job

1. Press **Menu** on the main screen
2. Press **Jobs**
3. Press **New** and enter a job name
4. Select a **Crop**, **Header**, and **Profile** from the drop-downs
5. Optionally select a **Field**
6. Press **Start Job**

Recording begins automatically when harvesting conditions are met (speed > 0.5 km/h and AOG sections are on).

Managing jobs

The jobs list shows all saved jobs with their status, date, area, and linked field.

Action	Description
Load	Make a saved job the active job (prompts if another job is already active)
Delete	Permanently remove a job and all its recorded data — cannot be undone

Only one job can be active at a time.

Auto-pause

The job automatically pauses when harvesting stops:

- Machine speed falls below 0.5 km/h, **or**
- AOG turns all header sections off (e.g. turning on headland, travelling over harvested ground)

When auto-paused the job name turns orange in the status bar. Recording resumes automatically when harvesting conditions return.

Manual pause and resume

Press **Pause** on the main screen to manually pause recording. Press **Start** to resume.

Stopping a job

Press **Stop** → confirm. The job is saved and closed. All totals are written to the database. The job remains in the list and can be viewed in the Job Report.

Resume on start

If **Resume Job on Start** is enabled in Settings, the active job is automatically reloaded when YieldFlo restarts. Useful if the PC is rebooted during a harvest day.

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4. Crops

Menu: Menu → Crops

Crops store the grain-specific parameters used in yield calculations and reporting.

Crop settings

Field	Description
Name	Crop name (e.g. Wheat, Corn, Canola)
Category	Grain type category
Test Weight	Standard bushel weight (lbs/bu) — used to convert mass to bushels
Market Moisture	Standard moisture for yield reporting (e.g. 14.5% for wheat)

Adding a crop

1. Press **New** and enter a crop name
2. Set the test weight and market moisture
3. Press **Save**

Select a crop from the list to edit it. At least one crop must exist at all times.

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5. Headers

Menu: Menu → Headers

Headers define the cutting width of the front attachment. Width is used to calculate the area harvested.

Header settings

Field	Description
Name	Header name (e.g. 30ft Draper, 8-row Corn Head)
Type	Header type category
Width	Cutting width in feet (Imperial) or metres (Metric)

Adding a header

1. Press **New** and enter a header name
2. Enter the cutting width
3. Press **Save**

At least one header must exist at all times.

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6. Profiles

Menu: Menu → Profiles

Profiles store combine-specific and calibration settings. Use a separate profile for each combine if running YieldFlo on multiple machines.

Profile settings

Field	Description
Name	Profile name (e.g. JD 9600, Case 2388)
Combine ID	Identifier for the module — must match the module configuration
Moisture offset	Fixed offset applied to all moisture readings
Moisture scale	Counts-per-percent for the moisture sensor calibration
Temperature offset	Temperature offset (°C)
Temperature scale	Counts-per-degree for the temperature sensor

Note: Moisture and temperature calibration values are set in **Menu → Moisture Cal** and saved automatically to the active profile.

Adding a profile

1. Press **New** and enter a profile name
2. Enter the Combine ID to match your module
3. Press **Save**

At least one profile must exist at all times.

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7. Fields

Menu: Menu → Fields

Fields are optional location labels that can be assigned to jobs for organisation and reporting.

Adding a field

1. Press **New** and enter a field name
2. Press **Save**

Fields are not required. A job can be created without a field selected.

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8. Yield Calibration

Menu: Menu → Yield Cal

Yield calibration corrects the elevator sensor reading to match the actual mass of grain harvested. Two parameters are set here:

- **Sensor Baseline** — the obstruction reading with paddles running but no grain (set once at installation)
- **Yield Factor** — a multiplier that scales the sensor reading to match a weighed reference

Setting the baseline

1. Start the clean-grain elevator with no grain
2. Open **Menu** → **Yield Cal**
3. When the reading is stable, press **Set Baseline**
4. Confirm the prompt

Tip: Run the empty elevator for at least 10 seconds before setting the baseline so the reading stabilises.

Processing delay

The processing delay accounts for the travel time of grain from the header to the elevator sensor. Set this value (in seconds) to match your combine. Typical range is 4–12 seconds.

Running a calibration pass

A calibration pass measures the actual mass of grain harvested during a known run, then adjusts the Yield Factor to match.

Preparation: Position a weigh wagon or grain cart to catch grain from the unloading auger and record start and end weights.

1. Open **Menu** → **Yield Cal**
2. Press **Start Run** — the app begins accumulating sensor data
3. Harvest a suitable area (at least one full tank is recommended)
4. Press **Stop Run**
5. Weigh the harvested grain
6. Enter the actual weight in the **Actual weight** field
7. Press **Apply Cal**

YieldFlo calculates a new Yield Factor and displays the result. Press **Save & Apply** to save the new factor to the active profile.

Manual factor adjustment

The Yield Factor can also be adjusted manually in the **Yield Factor** field. Increase to raise yield readings; decrease to lower them.

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9. Moisture Calibration

Menu: Menu → Moisture Cal

Moisture calibration maps the raw sensor counts from the capacitance moisture sensor to actual moisture percentage and grain temperature. Calibration values are stored per-profile.

The moisture reading is calculated as:

$$\text{Moisture (\%)} = \text{offset} + (\text{scale} \times \text{sensor count})$$

Moisture calibration procedure

1. Take a grain sample and measure moisture with a certified grain moisture meter
2. Open **Menu** → **Moisture Cal**
3. With the module connected and grain flowing, note the live sensor reading
4. Enter the meter reading in the **%** field
5. Adjust the **%/count** scale until the displayed reading matches
6. Press **Save**

Moisture calibration fields

Field	Description
% (saved to crop)	The reference moisture reading from an approved meter
%/count	The counts-per-percent scale factor (saved to the active profile)

Temperature calibration

Field	Description
°C	Reference temperature from a thermometer (saved to profile)
°C/count	Counts-per-degree scale factor

Note: Temperature calibration is optional. Temperature is used internally to compensate moisture readings and does not appear directly on the main screen.

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10. Yield Map

Menu: Menu → Yield Map

The yield map displays a colour-coded spatial plot of yield rate across the harvested area.

Data points are drawn as coloured circles at each GPS position. Colour ranges from blue (low yield) to red (high yield), with the **Low** and **High** legend values shown below the map.

Selecting a job

Use the job selector at the top of the map to choose which job to display. The map updates when a different job is selected.

Refreshing

Press **Refresh** to reload the map data. Useful if the map is open during an active job and you want to see the latest points.

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11. Job Report

Menu: Menu → Reports

The job report shows a summary of a completed or active job.

Report contents

Field	Description
Field	Field name assigned to the job
Area	Total area harvested (ac or ha)
Total	Total mass harvested (bu or tonnes)
Avg Yield	Average yield rate across all data points
Avg Moisture	Average grain moisture across all data points
Data Points	Number of recorded GPS data points

CSV export

Press **Export CSV** to save all raw data points for the job to a CSV file. The file is saved to the **Jobs** folder inside the YieldFlo application folder.

Column	Description
Timestamp	Date and time of the reading
Latitude	GPS latitude (decimal degrees)
Longitude	GPS longitude (decimal degrees)
Yield	Instantaneous yield rate
Moisture	Grain moisture (%)
Speed	Ground speed
Heading	GPS heading (degrees)
Area	Incremental area for this point

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12. Settings

Menu: Menu → Settings

Units

Setting	Options
Units	Imperial (bu/ac, mph, acres, lbs) or Metric (t/ha, km/h, ha, tonnes)

Changing units takes effect immediately. Historical data is stored in metric units internally and converted for display.

Theme

Setting	Options
Theme	Dark or Light

Module communication

Setting	Description
WiFi	Module communicates over UDP WiFi. Connect the PC to the module's WiFi access point or a shared network.
CAN Driver	Module communicates over CAN bus (Phase 2 hardware required).
COM Port	Serial port for CAN interface (CAN mode only).

Resume Job on Start

When enabled, YieldFlo automatically reloads the active job when the app starts. Useful if the PC is rebooted during a harvest day.

Saving settings

Press **Save & Apply** to save and apply all settings immediately.

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13. Language

Menu: Menu → Language

YieldFlo supports the following languages:

Code	Language
en	English
de	German
fr	French
hu	Hungarian
nl	Dutch
pl	Polish
ru	Russian
lt	Lithuanian

Changing language

1. Press **Menu → Language**
2. Select the desired language from the list
3. Press **Save & Restart**

YieldFlo will restart automatically with the new language applied.

Note: If AgOpenGPS is already installed with a language set, YieldFlo will automatically use the same language on first launch.

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14. Troubleshooting

GPS indicator stays red

- Confirm AgOpenGPS is running on the same PC or local network
- Check that both apps are on the same subnet
- In YieldFlo Settings, verify the network configuration

Module indicator stays off

- Check the module is powered and the WiFi or CAN connection is active
- In WiFi mode, confirm the PC is connected to the correct network
- Check the module LED for status indication
- Restart the module and wait 10–15 seconds for it to connect

Yield reads zero or very low

- Check the elevator optical sensor alignment and wiring
- Verify the Sensor Baseline is set correctly (set with elevator running empty)
- Confirm the processing delay is appropriate for your combine

Moisture reads incorrectly

- Verify the moisture sensor is installed and wired correctly
- Check that the moisture calibration (offset and scale) is set in the active profile
- Compare against a certified grain moisture meter and adjust the calibration

App crashes on start

- Check `YieldFlo_Crash.log` in the application folder for details

Job does not resume after restart

- Ensure **Resume Job on Start** is enabled in Settings
- The job must have been active (not stopped) when YieldFlo was closed

YieldFlo is designed for use with AgOpenGPS. It is not a certified weighing system and should not be used as the sole basis for grain settlement.